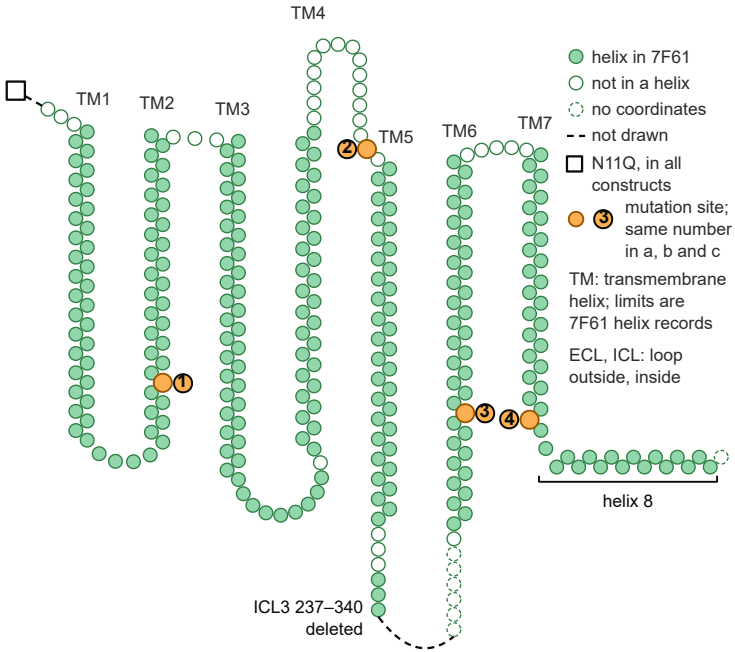
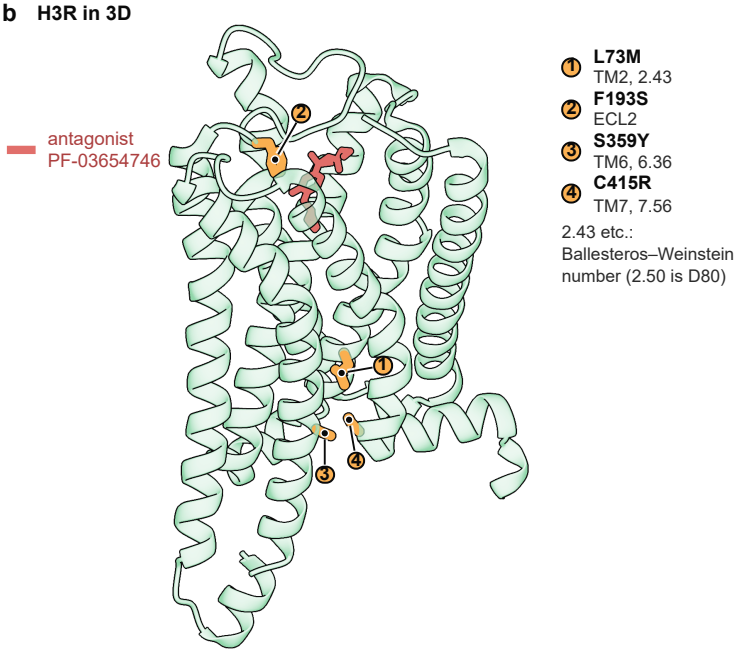
a **Four H3R mutations that restore signaling**

H3R: human histamine H3 receptor. Top is outside the cell.
Chain A of Protein Data Bank entry 7F61, residues 27–433 of 445.
Yeast construct H3R_i3d: N11Q, ICL3 237–340 deleted.

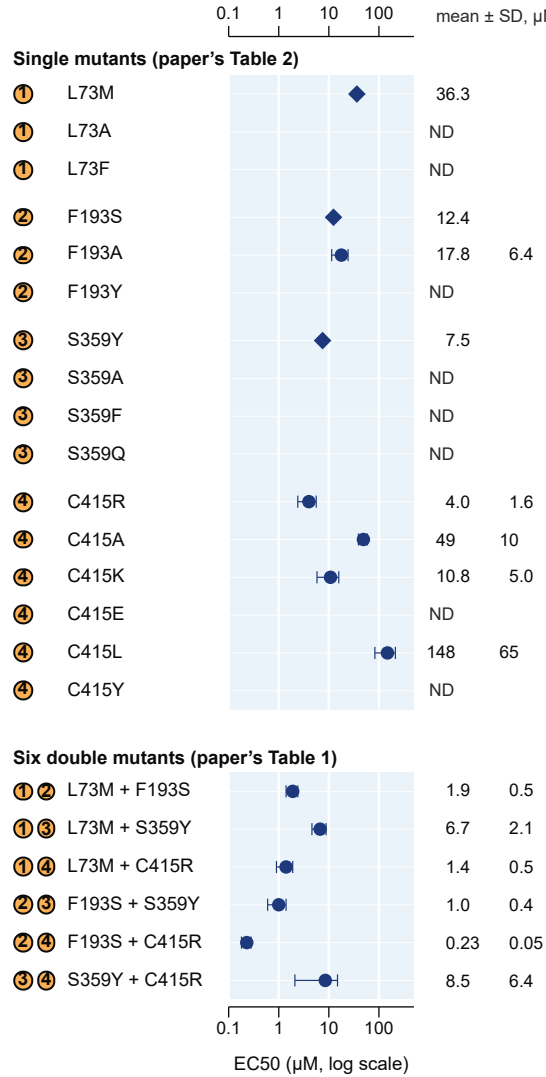


b **H3R in 3D**



c **Histamine EC50 of each H3R mutant**

EC50: histamine concentration, half-maximal response.
Lower EC50 = the mutant responds to less histamine.



ND: not determined; the paper's text calls some inactive, some weak.
C415R: 4.0 ± 1.6 here; 3.5 ± 1.5 in Table 1.
Doubles sorted by site number.
Panel b: tilted 30° towards the outside; heights depend on view.
ICL3 242–346: missing from the 7F61 model.
Not drawn: chain B (BRIL, a fusion protein for crystallisation), cholesterol, water.

Data: 7F61 coordinates and helix records, published EC50 values. Illustration: loops.

After Figs 1 and 4, Tables 1 and 2 of Nakajima et al., Protein Science 2026, doi:10.1002/pro.70408, CC BY 4.0. Data: PDB 7F61. Redesigned; values as published.